# EEG32 / PASSIVE WEARABLE ACQUISITION

32 simultaneous channels | shared REF + separate BIAS | 500 SPS/channel target

## PASSIVE ELECTRODES

32 signals + REF + BIAS
One physical electrode harness
J601 / 40-contact low-profile FFC

4.99k / 4.7nF input filters
No electrode power or buffers

PassiveInputs.kicad_sch

## ADS1299 #1 / U101

CH01-CH08 / AFE1.kicad_sch

Clock + BIAS master

## ADS1299 #2 / U201

CH09-CH16 / AFE2.kicad_sch

Shared clock / START

## ADS1299 #3 / U301

CH17-CH24 / AFE3.kicad_sch

Shared clock / START

## ADS1299 #4 / U401

CH25-CH32 / AFE4.kicad_sch

Shared clock / START

## nRF52832 / BLE

Four CS / DOUT / DRDY groups
Shared SCLK / MOSI / START
Latest MCU pin assignments kept
BLE RF network retained
MCU.kicad_sch

## REF / BIAS

REF -> RC -> four SRB1 pins
U101 BIAS -> R634 -> electrode
REF and BIAS remain separate.
ReferenceBias.kicad_sch

## POWER

P2: VBAT
U2: +3V3 digital
U3: +5VA analog
Common GND, four-layer PCB
Power.kicad_sch

Arrows show functional signal flow; named nets make electrical connections. Physical cable terminates at J601.
Active hub and electrode-module projects are archived reference designs, outside this passive assembly.

**ROUTED BENCH CHECKPOINT: manufacturing, measured performance and human-contact safety remain release gates.**

Passive bench prototype | protection and footprint release review pending

Sheet: /
File: EEG32.kicad_sch

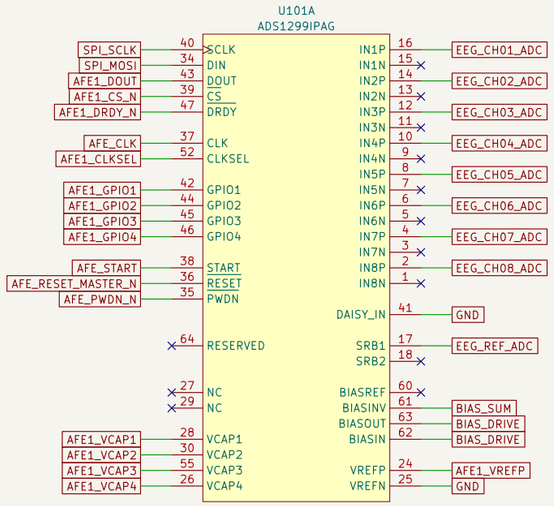
***Title: EEG32 passive | system overview***

| | | |
|---|---|---|
| Size: A3 | Date: 2026-10-01 | **Rev: 2.1-prefab-bench** |
| KiCad E.D.A. 10.0.5 | | Id: 1/9 |

# ADS1299 #1 / CH01–CH08

PassiveInputs.kicad_sch RC bank –> EEG_CH01_ADC ... EEG_CH08_ADC | shared SRB1 reference

## ADC INPUTS / CONTROL

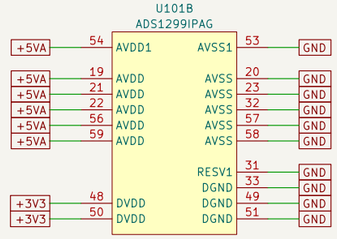


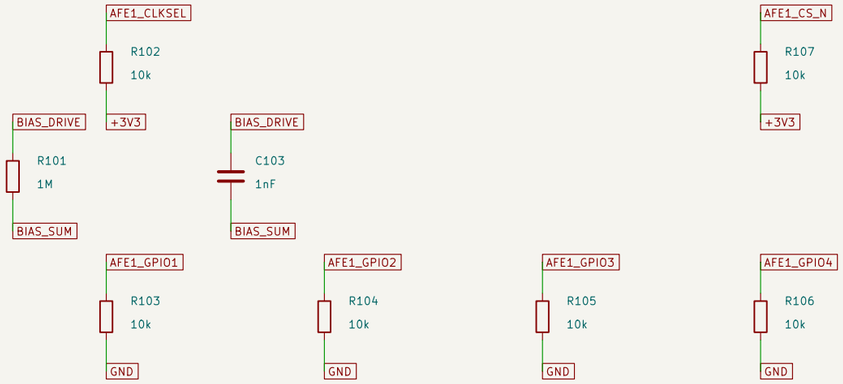

## PASSIVE INPUT BANK

CH01: J601.1 –> R601 –> IN1P / pin 16
CH02: J601.2 –> R602 –> IN2P / pin 14
CH03: J601.3 –> R603 –> IN3P / pin 12
CH04: J601.4 –> R604 –> IN4P / pin 10
CH05: J601.5 –> R605 –> IN5P / pin 8
CH06: J601.6 –> R606 –> IN6P / pin 6
CH07: J601.7 –> R607 –> IN7P / pin 4
CH08: J601.8 –> R608 –> IN8P / pin 2

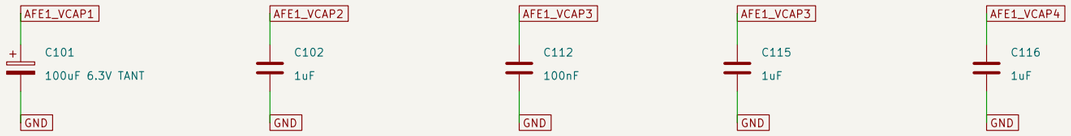
MASTER: CLKSEL pulled high; enable CLK_EN.
Only U101 drives BIAS; R101 || C103 is feedback.

## SUPPLIES / CONFIGURATION

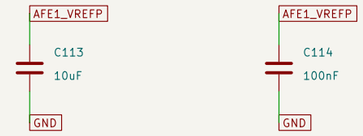


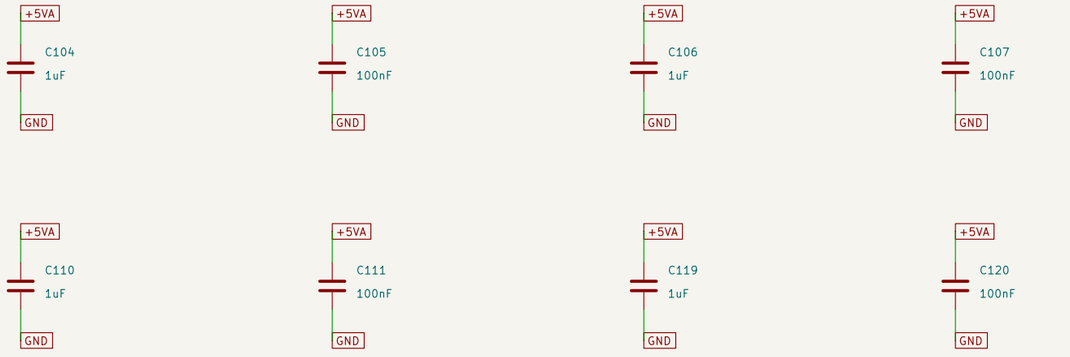

## VCAP SUPPORT



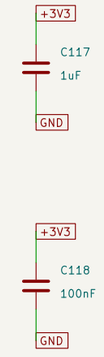

## LOCAL ADC REFERENCE




## ANALOG SUPPLY BYPASS / +5VA




## DIGITAL BYPASS




## PASSIVE COMMON–REFERENCE MODE

EEG_REF_ADC –> pin 17 / SRB1. Select MISC1.SRB1 in firmware; unused external INxN pins are NC.
ADC VREFP is local conversion reference, not the REF electrode. See ReferenceBias.kicad_sch.

Passive bench prototype | protection and footprint release review pending

Sheet: /CH01–CH08 / clock and BIAS master/
File: AFE1.kicad_sch

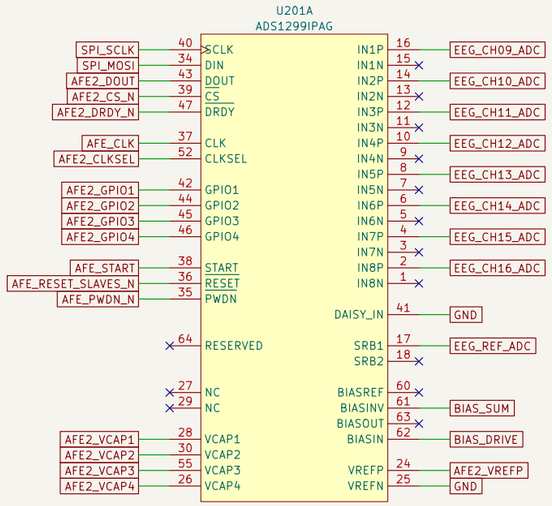
**Title: EEG32 passive | AFE1 | CH01–CH08**

| | | |
|---|---|---|
| Size: A3 | Date: 2026–10–01 | **Rev: 2.1–prefab–bench** |
| KiCad E.D.A. 10.0.5 | | Id: 2/9 |

# ADS1299 #2 / CH09–CH16

PassiveInputs.kicad_sch RC bank –> EEG_CH09_ADC ... EEG_CH16_ADC | shared SRB1 reference

## ADC INPUTS / CONTROL

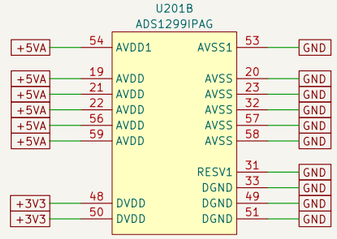


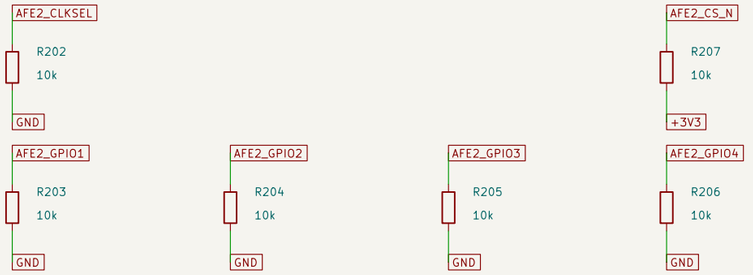

## PASSIVE INPUT BANK

CH09: J601.9 –> R609 –> IN1P / pin 16
CH10: J601.10 –> R610 –> IN2P / pin 14
CH11: J601.11 –> R611 –> IN3P / pin 12
CH12: J601.12 –> R612 –> IN4P / pin 10
CH13: J601.13 –> R613 –> IN5P / pin 8
CH14: J601.14 –> R614 –> IN6P / pin 6
CH15: J601.15 –> R615 –> IN7P / pin 4
CH16: J601.16 –> R616 –> IN8P / pin 2

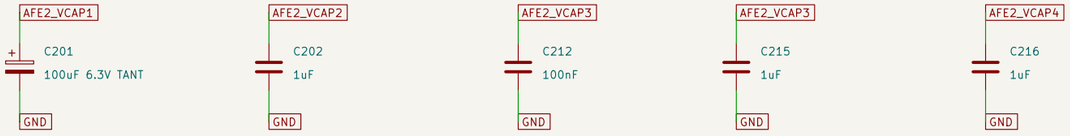
SLAVE: CLKSEL pulled low; AFE1 supplies CLK.
BIASOUT is intentionally NC; keep slave BIAS amplifier disabled.

## SUPPLIES / CONFIGURATION

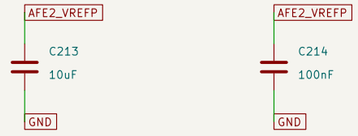


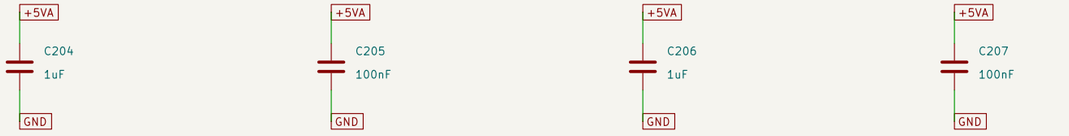

## VCAP SUPPORT



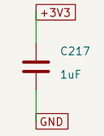

## LOCAL ADC REFERENCE




## ANALOG SUPPLY BYPASS / +5VA




## DIGITAL BYPASS




## PASSIVE COMMON–REFERENCE MODE

EEG_REF_ADC –> pin 17 / SRB1. Select MISC1.SRB1 in firmware; unused external INxN pins are NC.
ADC VREFP is local conversion reference, not the REF electrode. See ReferenceBias.kicad_sch.

Passive bench prototype | protection and footprint release review pending

Sheet: /CH09–CH16 / synchronous AFE/
File: AFE2.kicad_sch

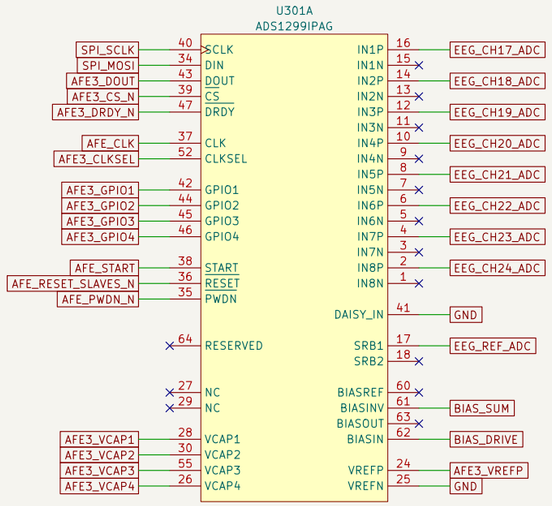
**Title: EEG32 passive | AFE2 | CH09–CH16**

| | | |
|---|---|---|
| Size: A3 | Date: 2026–10–01 | **Rev: 2.1–prefab–bench** |
| KiCad E.D.A. 10.0.5 | | Id: 3/9 |

# ADS1299 #3 / CH17–CH24

PassiveInputs.kicad_sch RC bank –> EEG_CH17_ADC ... EEG_CH24_ADC | shared SRB1 reference

## ADC INPUTS / CONTROL

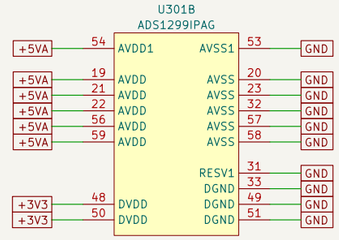


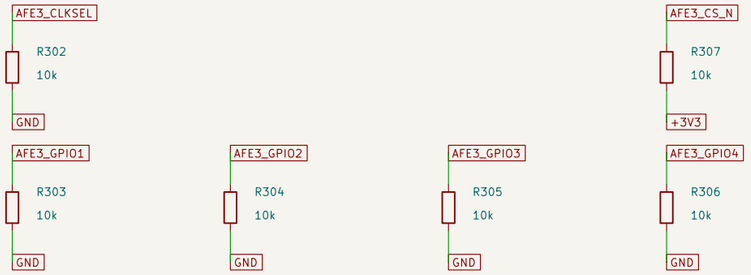

## PASSIVE INPUT BANK

CH17: J601.17 –> R617 –> IN1P / pin 16
CH18: J601.18 –> R618 –> IN2P / pin 14
CH19: J601.19 –> R619 –> IN3P / pin 12
CH20: J601.20 –> R620 –> IN4P / pin 10
CH21: J601.21 –> R621 –> IN5P / pin 8
CH22: J601.22 –> R622 –> IN6P / pin 6
CH23: J601.23 –> R623 –> IN7P / pin 4
CH24: J601.24 –> R624 –> IN8P / pin 2

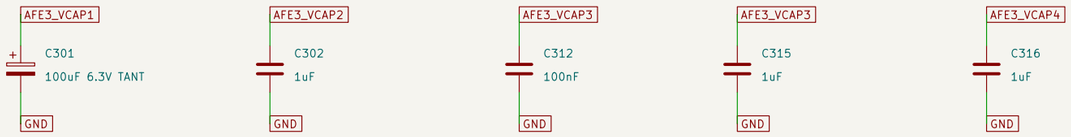
SLAVE: CLKSEL pulled low; AFE1 supplies CLK.
BIASOUT is intentionally NC; keep slave BIAS amplifier disabled.

## SUPPLIES / CONFIGURATION

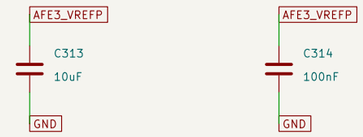


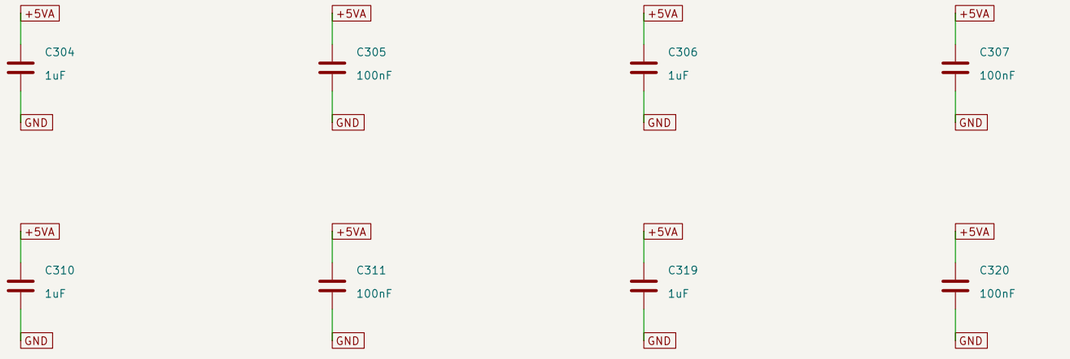

## VCAP SUPPORT



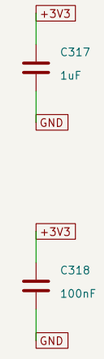

## LOCAL ADC REFERENCE




## ANALOG SUPPLY BYPASS / +5VA




## DIGITAL BYPASS




## PASSIVE COMMON–REFERENCE MODE

EEG_REF_ADC –> pin 17 / SRB1. Select MISC1.SRB1 in firmware; unused external INxN pins are NC.
ADC VREFP is local conversion reference, not the REF electrode. See ReferenceBias.kicad_sch.

Passive bench prototype | protection and footprint release review pending

Sheet: /CH17–CH24 / synchronous AFE/
File: AFE3.kicad_sch

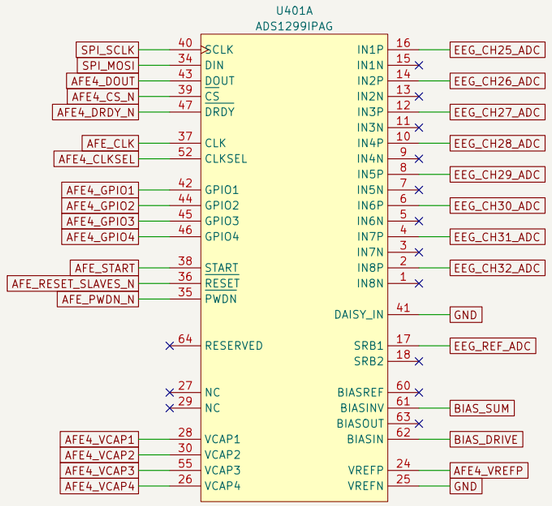
**Title: EEG32 passive | AFE3 | CH17–CH24**

| | | |
|---|---|---|
| Size: A3 | Date: 2026–10–01 | **Rev: 2.1–prefab–bench** |
| KiCad E.D.A. 10.0.5 | | Id: 4/9 |

# ADS1299 #4 / CH25–CH32

PassiveInputs.kicad_sch RC bank –> EEG_CH25_ADC ... EEG_CH32_ADC | shared SRB1 reference

## ADC INPUTS / CONTROL

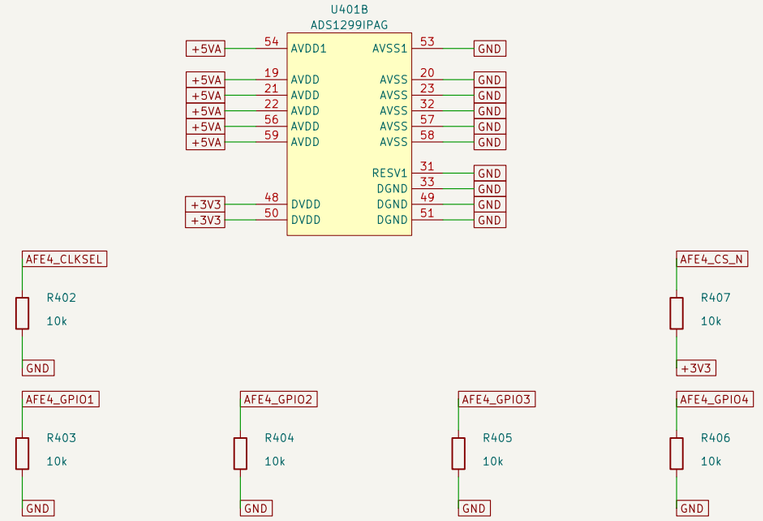



## PASSIVE INPUT BANK

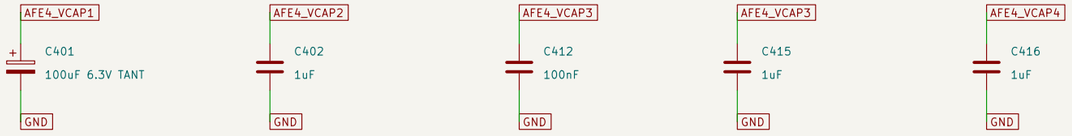
CH25: J601.25 –> R625 –> IN1P / pin 16
CH26: J601.26 –> R626 –> IN2P / pin 14
CH27: J601.27 –> R627 –> IN3P / pin 12
CH28: J601.28 –> R628 –> IN4P / pin 10
CH29: J601.29 –> R629 –> IN5P / pin 8
CH30: J601.30 –> R630 –> IN6P / pin 6
CH31: J601.31 –> R631 –> IN7P / pin 4
CH32: J601.32 –> R632 –> IN8P / pin 2

SLAVE: CLKSEL pulled low; AFE1 supplies CLK.
BIASOUT is intentionally NC; keep slave BIAS amplifier disabled.

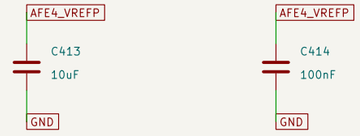
## SUPPLIES / CONFIGURATION



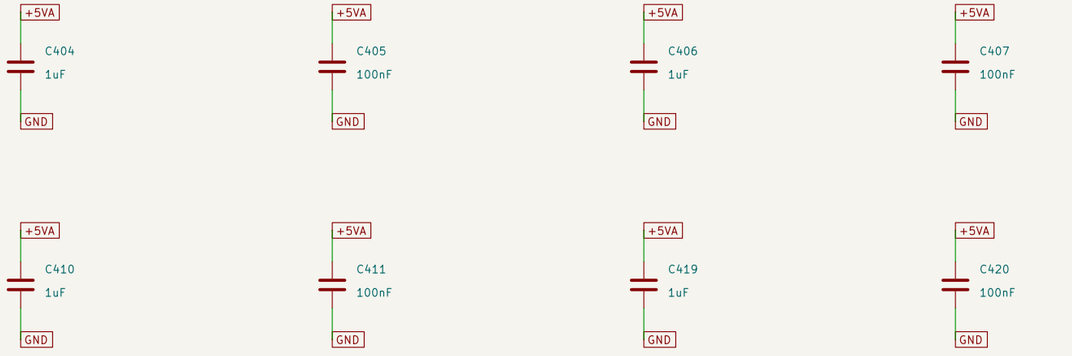

## VCAP SUPPORT



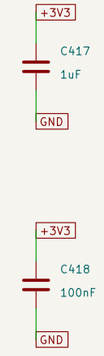

## LOCAL ADC REFERENCE




## ANALOG SUPPLY BYPASS / +5VA




## DIGITAL BYPASS




## PASSIVE COMMON–REFERENCE MODE

EEG_REF_ADC –> pin 17 / SRB1. Select MISC1.SRB1 in firmware; unused external INxN pins are NC.
ADC VREFP is local conversion reference, not the REF electrode. See ReferenceBias.kicad_sch.

Passive bench prototype | protection and footprint release review pending

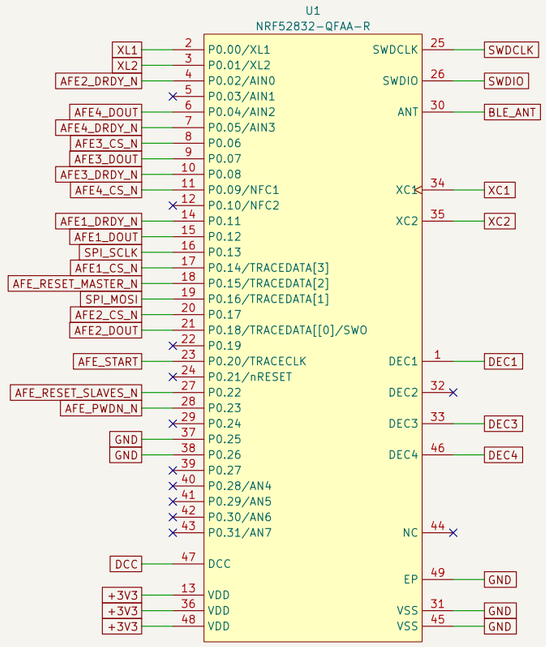
Sheet: /CH25–CH32 / synchronous AFE/
File: AFE4.kicad_sch

**Title: EEG32 passive | AFE4 | CH25–CH32**

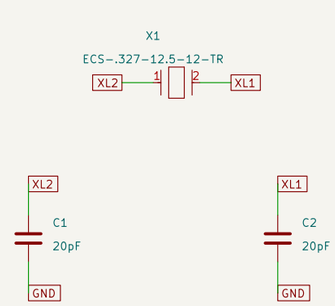
Size: A3 | Date: 2026–10–01 | **Rev: 2.1–prefab–bench**
KiCad E.D.A. 10.0.5 | Id: 5/9

# nRF52832 / ACQUISITION + BLE

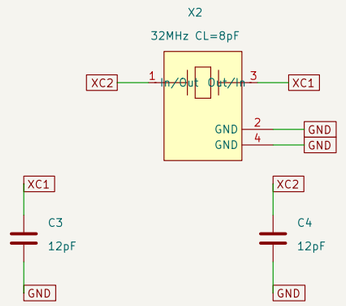
U1: four independent AFE data/ready/select groups; shared SPI clock and MOSI



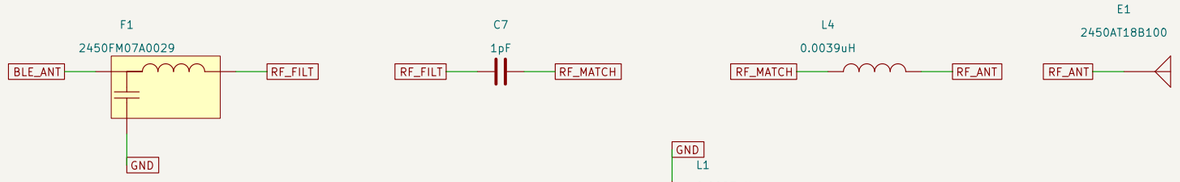

## LOW-FREQUENCY CLOCK



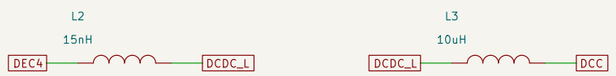

## 32 MHz CLOCK




## RF MATCHING / ANTENNA



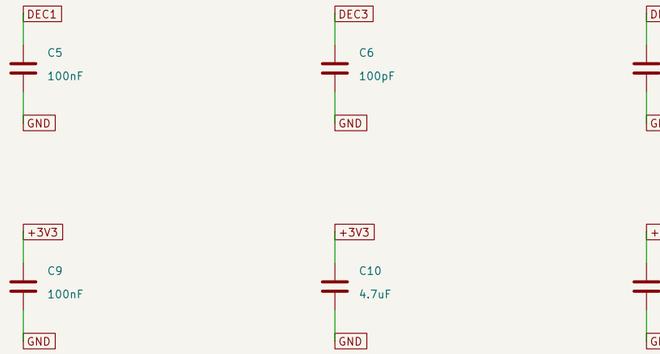

## MCU DC/DC SUPPORT



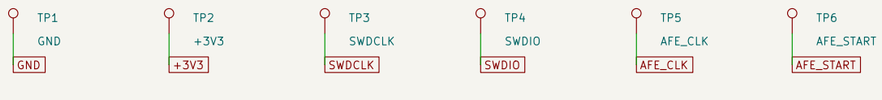

## CONTROL DEFAULTS




## SUPPLY / DEC BYPASS

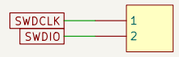



## DEBUG / TIMING TEST POINTS




## P1

Header 2




AFE4_CS_N = U1.11 / P0.09. Configure NFC pin as GPIO.
Shared START; separate master/slave reset; 500 SPS/channel target.
RF and oscillator values/geometry retained; only unused GPIO37/38 grounded.

ERRATUM 138: U1.37/P0.25 and U1.38/P0.26 are unused and tied to exposed VSS pad.
Do not configure either as a driven-high output or peripheral pin. Verify silicon lot/build code before release.

Passive bench prototype | protection and footprint release review pending

Sheet: /nRF52832 / BLE / synchronized acquisition control/
File: MCU.kicad_sch

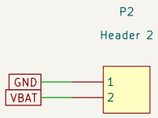
**Title: EEG32 passive | MCU**

| | | |
|---|---|---|
| Size: A3 | Date: 2026-10-01 | **Rev: 2.1-prefab-bench** |
| KiCad E.D.A. 10.0.5 | | Id: 6/9 |

# POWER SUPPLIES / ANALOG ACCESS

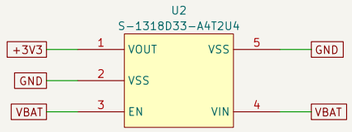
VBAT independently feeds the +3V3 LDO and +5VA boost. Common board GND is the supply return.

## BATTERY INPUT + 3.3 V DIGITAL RAIL

















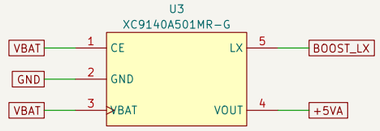


U2 output supplies the MCU and ADS digital pins.
Headroom at minimum battery voltage needs measurement.

P2.2 = VBAT; P2.1 = GND.
Battery protection is external / unqualified.




## 5 V ANALOG RAIL







U3 supplies the four ADS1299 analog sections.
No electrode power is exported.
Capacity, startup and ripple require load testing.

## TEST ACCESS / REF IS NOT VREFP OR GND













TP7: EEG_REF_ADC —> all SRB1. TP8: BIAS_DRIVE before the electrode series resistor.
Capacitor values are preserved. MPNs specified in 2.1; qualify effective capacitance before release.

Passive bench prototype | protection and footprint release review pending

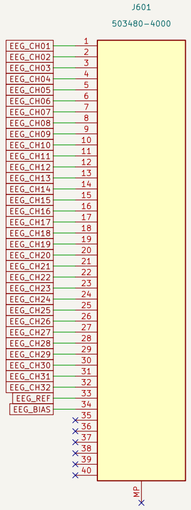
Sheet: /Battery input / 3.3V LDO / 5V boost / REF and BIAS access/
File: Power.kicad_sch

**Title: EEG32 passive | Power**

| Size: A3 | Date: 2026-10-01 | **Rev: 2.1-prefab-bench** |
|---|---|---|
| KiCad E.D.A. 10.0.5 | | Id: 7/9 |

# PASSIVE ELECTRODE INTERFACE / ONE LOW-PROFILE FFC

J601: 32 signals + REF + BIAS | no active modules, external hub or electrode power




Molex 503480-4000 / 1.0mm height
40-way, 0.50mm pitch, dual-contact FFC
0.30 +/-0.03mm mating thickness
Pins 35-40: NC, no electrode connection.
MP pads: mechanical retention, NC.
External cable strain relief required.
Cable: Molex 0151660431, Type A, 102mm.
See prefab-20261001/FFC-MATING.md.

## FILTER BASIS / RELEASE LIMITS

TI SLAU443B, Fig. 59: 4.99k + 4.7nF
to ground per input. Same RC on REF.
Nominal pole: 6.79 kHz for zero source Z.
These are EMI filters, not 250 Hz
anti-alias filters or patient protection.
Electrode impedance changes bandwidth.
Validate leakage, ESD, powered-off faults,
noise and common-mode rejection.
No external amplifier is added.

J601.33 -> ReferenceBias / EEG_REF
J601.34 <- ReferenceBias / EEG_BIAS
REF and BIAS are separate electrodes.
No rail or GND contact goes to the scalp.

### AFE1 / CH01-CH08

R601 4.99k 1% — EEG_CH01 — C601 4.7nF 5% C0G 50V — GND — EEG_CH01_ADC
J601.1 -> U101.16 / IN1P

R602 4.99k 1% — EEG_CH02 — C602 4.7nF 5% C0G 50V — GND — EEG_CH02_ADC
J601.2 -> U101.14 / IN2P

R603 4.99k 1% — EEG_CH03 — C603 4.7nF 5% C0G 50V — GND — EEG_CH03_ADC
J601.3 -> U101.12 / IN3P

R604 4.99k 1% — EEG_CH04 — C604 4.7nF 5% C0G 50V — GND — EEG_CH04_ADC
J601.4 -> U101.10 / IN4P

R605 4.99k 1% — EEG_CH05 — C605 4.7nF 5% C0G 50V — GND — EEG_CH05_ADC
J601.5 -> U101.8 / IN5P

R606 4.99k 1% — EEG_CH06 — C606 4.7nF 5% C0G 50V — GND — EEG_CH06_ADC
J601.6 -> U101.6 / IN6P

R607 4.99k 1% — EEG_CH07 — C607 4.7nF 5% C0G 50V — GND — EEG_CH07_ADC
J601.7 -> U101.4 / IN7P

R608 4.99k 1% — EEG_CH08 — C608 4.7nF 5% C0G 50V — GND — EEG_CH08_ADC
J601.8 -> U101.2 / IN8P

### AFE2 / CH09-CH16

R609 4.99k 1% — EEG_CH09 — C609 4.7nF 5% C0G 50V — GND — EEG_CH09_ADC
J601.9 -> U201.16 / IN1P

R610 4.99k 1% — EEG_CH10 — C610 4.7nF 5% C0G 50V — GND — EEG_CH10_ADC
J601.10 -> U201.14 / IN2P

R611 4.99k 1% — EEG_CH11 — C611 4.7nF 5% C0G 50V — GND — EEG_CH11_ADC
J601.11 -> U201.12 / IN3P

R612 4.99k 1% — EEG_CH12 — C612 4.7nF 5% C0G 50V — GND — EEG_CH12_ADC
J601.12 -> U201.10 / IN4P

R613 4.99k 1% — EEG_CH13 — C613 4.7nF 5% C0G 50V — GND — EEG_CH13_ADC
J601.13 -> U201.8 / IN5P

R614 4.99k 1% — EEG_CH14 — C614 4.7nF 5% C0G 50V — GND — EEG_CH14_ADC
J601.14 -> U201.6 / IN6P

R615 4.99k 1% — EEG_CH15 — C615 4.7nF 5% C0G 50V — GND — EEG_CH15_ADC
J601.15 -> U201.4 / IN7P

R616 4.99k 1% — EEG_CH16 — C616 4.7nF 5% C0G 50V — GND — EEG_CH16_ADC
J601.16 -> U201.2 / IN8P

### AFE3 / CH17-CH24

R617 4.99k 1% — EEG_CH17 — C617 4.7nF 5% C0G 50V — GND — EEG_CH17_ADC
J601.17 -> U301.16 / IN1P

R618 4.99k 1% — EEG_CH18 — C618 4.7nF 5% C0G 50V — GND — EEG_CH18_ADC
J601.18 -> U301.14 / IN2P

R619 4.99k 1% — EEG_CH19 — C619 4.7nF 5% C0G 50V — GND — EEG_CH19_ADC
J601.19 -> U301.12 / IN3P

R620 4.99k 1% — EEG_CH20 — C620 4.7nF 5% C0G 50V — GND — EEG_CH20_ADC
J601.20 -> U301.10 / IN4P

R621 4.99k 1% — EEG_CH21 — C621 4.7nF 5% C0G 50V — GND — EEG_CH21_ADC
J601.21 -> U301.8 / IN5P

R622 4.99k 1% — EEG_CH22 — C622 4.7nF 5% C0G 50V — GND — EEG_CH22_ADC
J601.22 -> U301.6 / IN6P

R623 4.99k 1% — EEG_CH23 — C623 4.7nF 5% C0G 50V — GND — EEG_CH23_ADC
J601.23 -> U301.4 / IN7P

R624 4.99k 1% — EEG_CH24 — C624 4.7nF 5% C0G 50V — GND — EEG_CH24_ADC
J601.24 -> U301.2 / IN8P

### AFE4 / CH25-CH32

R625 4.99k 1% — EEG_CH25 — C625 4.7nF 5% C0G 50V — GND — EEG_CH25_ADC
J601.25 -> U401.16 / IN1P

R626 4.99k 1% — EEG_CH26 — C626 4.7nF 5% C0G 50V — GND — EEG_CH26_ADC
J601.26 -> U401.14 / IN2P

R627 4.99k 1% — EEG_CH27 — C627 4.7nF 5% C0G 50V — GND — EEG_CH27_ADC
J601.27 -> U401.12 / IN3P

R628 4.99k 1% — EEG_CH28 — C628 4.7nF 5% C0G 50V — GND — EEG_CH28_ADC
J601.28 -> U401.10 / IN4P

R629 4.99k 1% — EEG_CH29 — C629 4.7nF 5% C0G 50V — GND — EEG_CH29_ADC
J601.29 -> U401.8 / IN5P

R630 4.99k 1% — EEG_CH30 — C630 4.7nF 5% C0G 50V — GND — EEG_CH30_ADC
J601.30 -> U401.6 / IN6P

R631 4.99k 1% — EEG_CH31 — C631 4.7nF 5% C0G 50V — GND — EEG_CH31_ADC
J601.31 -> U401.4 / IN7P

R632 4.99k 1% — EEG_CH32 — C632 4.7nF 5% C0G 50V — GND — EEG_CH32_ADC
J601.32 -> U401.2 / IN8P

Wet Ag/AgCl: 5-10 kohm target | 0.5-100 Hz EEG / 500 SPS | TI RC retained: ADS sinc3 response and bench validation still required

Bench prototype; not released for fabrication or human contact

Sheet: /PassiveInputs/
File: PassiveInputs.kicad_sch

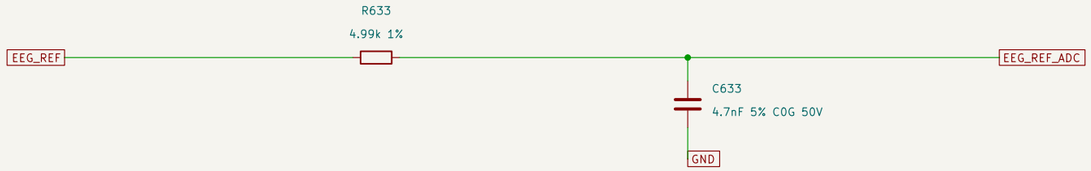
**Title: EEG32 passive | PassiveInputs**

Size: A2 | Date: 2026-10-01 | **Rev: 2.1-prefab-bench**
KiCad E.D.A. 10.0.5 | Id: 8/9

# SHARED REF / SEPARATE BIAS

Passive common-reference montage. No local electrode buffers; no REF/BIAS short.

## J601.33 / passive REF




U101/U201/U301/U401: pin 17 / SRB1
MISC1.SRB1 = 1; SRB2 remains unused.
TP7 measures the filtered shared reference.

Unbuffered shared REF retained: validate aggregate input loading and lead-off configuration for 32 channels.
One common REF filter: do not add 32 parallel filter capacitors to this electrode.
TI EVM default directly selects REF_ELEC to SRB1 (JP8); its optional buffer is not populated here.




U101 BIASOUT, pin 63
Existing 1M || 1nF feedback
remains before R634.

J601.34 / separate BIAS electrode
4.99k is TI EVM R10 value; it is not
a validated patient-current limiter.

## FIRMWARE + VALIDATION

Enable only U101 BIAS amplifier; slave BIASOUT pins remain NC. Preserve shared BIAS_SUM and BIASIN.
Select BIAS sense channels deliberately. Validate stability with electrode/cable impedance and disconnect faults.
At +5VA, internal BIASREF targets mid-supply. REF is neither GND nor the ADS VREFP conversion reference.
Common clock + shared START synchronize all four devices. Configure identical 500 SPS rates.
Lead-off excitation and gain/DC-offset range require bench verification with passive electrode models.

**PROTECTION RELEASE HOLD: RC networks reduce input current/EMI but do not establish ESD, defibrillator, isolation, leakage or single-fault safety. No human connection until supervisor/electrical safety validation.**

Bench prototype; not released for fabrication or human contact

Sheet: /ReferenceBias/
File: ReferenceBias.kicad_sch

***Title: EEG32 passive | ReferenceBias***

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|---|---|---|
| Size: A3 | Date: 2026-10-01 | **Rev: 2.1-prefab-bench** |
| KiCad E.D.A. 10.0.5 | | Id: 9/9 |